

2.5. Component Interfaces

2.5.1. Application Server - exposed API

For RESTful APIs exposed by the Application Server, parameters and return objects types won't be specified, as they will all be encoded into JSON structures according to REST standards.

AuthenticationManager

Request	GET /api/auth/start_stud_ver
Parameters	student information
Description	Redirects the student to the verification platform (status code 302).

Table 2.1: Authentication Manager REST request - start student verification

Request	GET /api/auth/verify_student
Parameters	institutional credentials
Description	Hands over the student enrollment verification to the Student Verification Platform. Redirects the student to the registration page (status code 302), providing a success response to the Web Browser in the case of successful verification or an error response in the case of failed verification.

Table 2.2: Authentication Manager REST request - student verification

Request	POST /api/auth/reg_student
Parameters	student information
Description	Hands over the student registration to the Data Access Manager. Provides a success response to the Web Browser (status code 201) if the inserted e-mail address is unique, or an error response (status code 409) if the address is already used.

Table 2.3: Authentication Manager REST request - student registration

Request	POST /api/auth/reg_company
Parameters	company information
Description	Handles the company registration process. Provides a success response to the Web Browser (status code 201) if the inserted e-mail address is unique and an error response if the address is already used (status code 409) or in the case of failed company verification (status code 404).

Table 2.4: Authentication Manager REST request - company registration

Request	POST /api/auth/login
Parameters	credentials
Description	Hands over user login to the Data Access Manager. If the inserted credentials are valid, it redirects the user to their homepage by providing a success response to the Web Browser (status code 200), otherwise it provides an error response (status code 401).

Table 2.5: Authentication Manager REST request - user login**StudentManager**

Request	POST /api/student/CV
Parameters	CV, userID
Description	Handles the CV upload process by interacting with the Data Access Manager. Returns a success response to the Web Browser in the case of successful upload (status code 201) or an error response if the CV document format is not supported (status code 415).

Table 2.6: Student Manager REST request - CV upload

Request	DELETE /api/student/CV
Parameters	CVID
Description	Handles the CV removal process by interacting with the Data Access Manager. Returns a success response to the Web Browser (status code 204).

Table 2.7: Student Manager REST request - CV removal

Request	GET /api/student/CV/recommendations
Parameters	CVID
Description	Retrieves the set of recommended internships for a specific student by interacting with the Data Access Manager. If there are any recommendations, it returns them to the Web Browser (status code 201), otherwise it returns a warning response (status code 404).

Table 2.8: Student Manager REST request - recommended internships retrieval

Request	POST /api/student/CV/selection
Parameters	internshipID, CVID
Description	Hands over the internship selection process to the Matching Manager. Returns a success response to the Web Browser (status code 201).

Table 2.9: Student Manager REST request - internship selection

Request	GET /api/student/CV/interested
Parameters	CVID
Description	By interacting with the Matching Manager, it retrieves the set of internships for which companies have selected the student's CV and returns them to the Web Browser (status code 200).

Table 2.10: Student Manager REST request - interested companies retrieval

Request	POST /api/student/CV/accept
Parameters	internshipID, CVID
Description	Hands over the internship acceptance process to the Matching Manager. Returns a success response to the Web Browser (status code 201).

Table 2.11: Student Manager REST request - match creation

Request	GET /api/student/calendar
Parameters	studentID
Description	Retrieves the set of planned interviews for a student by interacting with the Selection Manager. Returns interviews information to the Web Browser (status code 200).

Table 2.12: Student Manager REST request - calendar retrieval

Request	POST /api/student/CV/sel_quest
Parameters	questionnaire
Description	Hands over the evaluation of a selection process questionnaire to the Selection Manager and returns a success response to the Web Browser (status code 202).

Table 2.13: Student Manager REST request - selection questionnaire compilation

Request	GET /api/student/sat_questions
Parameters	studentID
Description	Retrieves the set of student satisfaction questionnaires by interacting with the Recommendation Manager. Returns the questionnaires to the Web Browser (status code 200).

Table 2.14: Student Manager REST request - satisfaction questionnaires retrieval

Request	POST /api/student/CV/sat_quest
Parameters	questionnaire, match
Description	Hands over the evaluation of a student satisfaction questionnaire to the Selection Manager and returns a success response to the Web Browser (status code 202).

Table 2.15: Student Manager REST request - satisfaction questionnaire compilation

Request	GET /api/student/messages
Parameters	studentID
Description	Retrieves the messaging sessions related to student's active internships by interacting with the Messaging Manager. Returns the messages to the Web Browser (status code 200).

Table 2.16: Student Manager REST request - messages retrieval

Request	POST /api/student/messages
Parameters	mex, internshipID, studentID
Description	After the student sends a new message to a company, it hands over the process of messaging session update to the Messaging Manager. Returns a success response to the Web Browser (status code 201).

Table 2.17: Student Manager REST request - message communication

CompanyManager

Request	POST /api/company/internship
Parameters	title, document, companyID
Description	Handles the internship publication process by interacting with the Data Access Manager. Returns a success response to the Web Browser in the case of successful publication (status code 201) or an error response if the internship document format is not supported (status code 415).

Table 2.18: Company Manager REST request - internship publication

Request	GET /api/company/internship
Parameters	internshipID
Description	By interacting with the Data Access Manager, it retrieves all information related to a published internship (recommended CVs, interested students, completed matches...). Returns the obtained data to the Web Browser (status code 200).

Table 2.19: Company Manager REST request - internship section retrieval

Request	DELETE /api/company/internship
Parameters	internshipID
Description	Handles the internship removal process by interacting with the Data Access Manager. Returns a success response to the Web Browser (status code 204).

Table 2.20: Company Manager REST request - internship removal

Request	POST /api/company/internship/selection
Parameters	internshipID, CVID
Description	Hands over the CV selection process to the Matching Manager. Returns a success response to the Web Browser (status code 201).

Table 2.21: Company Manager REST request - CV selection

Request	POST /api/company/internship/accept
Parameters	internshipID, CVID
Description	Hands over the CV acceptance process to the Matching Manager. Returns a success response to the Web Browser (status code 201).

Table 2.22: Company Manager REST request - match creation

Request	POST /api/company/internship/interview
Parameters	matchID, interview, questionnaire
Description	Hands over the interview set up process to the Selection Manager. If the specified date and time are free, it returns a success response to the Web Browser (status code 201), otherwise it returns an error response (status code 409).

Table 2.23: Company Manager REST request - interview set up

Request	POST /api/company/internship/admit
Parameters	matchID
Description	Hands over the admission of a student to the Selection Manager. Returns a success response to the Web Browser (status code 201).

Table 2.24: Company Manager REST request - student admission

Request	POST /api/company/internship/reject
Parameters	matchID
Description	Hands over the rejection of a student to the Selection Manager. Returns a success response to the Web Browser (status code 201).

Table 2.25: Company Manager REST request - student rejection

Request	POST /api/company/internship/sat_quest
Parameters	questionnaire, match
Description	Hands over the evaluation of a company satisfaction questionnaire to the Selection Manager and returns a success response to the Web Browser (status code 202).

Table 2.26: Company Manager REST request - satisfaction questionnaire compilation

Request	GET /api/company/internship/messages
Parameters	internshipID
Description	Retrieves the messaging sessions related to an internship by interacting with the Messaging Manager. Returns the messages to the Web Browser (status code 200).

Table 2.27: Company Manager REST request - messages retrieval

Request	POST /api/company/internship/messages
Parameters	mex, studentID, internshipID
Description	After a company sends a new message to a student, it hands over the process of messaging session update to the Messaging Manager. Returns a success response to the Web Browser (status code 201).

Table 2.28: Company Manager REST request - message communication

StudentVerificationPlatform

Request	GET /stud_ver_api/start_verification
Parameters	university
Description	Sets up the enrollment verification using student's university information. Returns a success response to the Authentication Manager (status code 200).

Table 2.29: Student Verification Platform REST request - redirect to platform

Request	GET /stud_ver_api/verify
Parameters	institutional credentials
Description	Verifies if a student is correctly enrolled in the specified university through their institutional credentials. Returns a success response to the Authentication Manager in the case of successful verification (status code 200) or an error message in the case of failed verification (status code 404).

Table 2.30: Student Verification Platform REST request - student verification**CompanyVerificationPlatform**

Request	GET /comp_ver_api/verify
Parameters	business registration number
Description	Verifies the identity of a company through its business registration number. Returns a success response to the Authentication Manager in the case of successful verification (status code 200) or an error message in the case of failed verification (status code 404).

Table 2.31: Company Verification Platform REST request - company verification

2.5.2. Application Server - internal interfaces

RecommendationManager

- `find_keywords(CV CV)`

Extracts keywords from a CV document and hands over recommendations update to the Data Access Manager. Returns an *Object* containing new recommendations (internships and related companies) for the corresponding student.

- `remove_recomm_int(String CVID)`

By interacting with the Data Access Manager, removes from all internships for which the canceled CV has been recommended all the related recommendations.

- `find_keywords(Internship internship)`

Extracts keywords from an internship document and hands over recommendations update to the Data Access Manager. Returns an *Object* containing new recommendations (students) for the corresponding company.

- `remove_recomm_int(String internshipID)`

By interacting with the Data Access Manager, removes from all CVs for which the canceled internship has been recommended all the related recommendations.

- `create_match_requests(Match match)`

Creates two satisfaction questionnaires for a completed matchmaking process (one for the student and one for the company).

- `create_int_requests(Match match)`

Creates two satisfaction questionnaires for an activated internship (one for the student and one for the company).

- `get_sat_requests(String studentID)`

Retrieves the set of satisfaction questionnaires for a student by interacting with the Data Access Manager. Returns a *List<Questionnaire>* object to the Student Manager.

- `evaluate_quest(Questionnaire questionnaire, Match match)`

Evaluates a satisfaction questionnaire for a matchmaking process or an active internship. Returns an *Object* containing information about students and internships

(and related companies) whose recommendations have been updated after the questionnaire evaluation.

MatchingManager

- `select_CV(String CVID, String internshipID)`

Hands over the CV selection process for the provided internship to the Data Access Manager. Returns an *Object* containing information about related student and internship.

- `select_int(String internshipID, String CVID)`

Hands over the internship selection process to the Data Access Manager. Returns an *Object* containing information about related company and CV.

- `interested_companies(String CVID)`

By interacting with the Data Access Manager, it retrieves the set of internships for which companies have selected the student's CV. Returns a *List<Internship>* object to the Student Manager.

- `accept_int(String internshipID, String CVID)`

Hands over the internship acceptance process to the Data Access Manager. Returns an *Object* containing information about the created match and the related company and student.

- `accept_CV(String internshipID, String CVID)`

Hands over the CV acceptance process for the provided internship to the Data Access Manager. Returns an *Object* containing information about the created match and the related company and student.

SelectionManager

- `setup_interview(String matchID, Interview interview, Questionnaire quest)`

Hands over the process of interview set up to the Data Access Manager. If the specified date and time are free, it returns a *Student* object (related to the interviewed student) to the Company Manager, otherwise it throws an *Exception*.

- `get_calendar(String studentID)`

Retrieves the set of planned interviews for a student by interacting with the Data Access Manager. Returns a *List<Interview>* object to the Student Manager.

- `grade_quest(Questionnaire questionnaire)`

Grades a selection process questionnaire filled out by a student and hands over the suitability rank update (for the company that created the questionnaire) to the Data Access Manager. Returns an *Object* containing information about the involved student and company.

- `admit_student(String matchID)`

Hands over the admission of a student to the Data Access Manager. Returns an *Object* containing information about the activated internship and the involved student and company.

- `reject_student(String matchID)`

Hands over the rejection of a student to the Data Access Manager. Returns an *Object* containing information about the related match and the involved student.

MessagingManager

- `get_stud_messages(String studentID)`

Retrieves the messaging sessions related to student's active internships by interacting with the Data Access Manager. Returns a *List<Message>* object to the Student Manager.

- `send_stud_mex(Message mex, String studentID, String internshipID)`

After the student sends a new message to a company, it hands over the process of messaging session update to the Data Access Manager. Returns an *Object* containing information about the related active internship and the involved student and company.

- `get_int_messages(String internshipID)`

Retrieves the messaging sessions related to an internship by interacting with the Messaging Manager. Returns a *List<Message>* to the Company Manager.

- `send_comp_mex(Message mex, String studentID, String internshipID)`

After the company sends a new message to a student, it hands over the process of messaging session update to the Data Access Manager. Returns an *Object* containing information about the related active internship and the involved student and company.

NotificationManager

- `recomm_notifications(List<Company> companies, List<Internship> internships)`
Notifies the provided companies of new recommended CVs for some of their internships.
- `recomm_notifications(List<Student> students)`
Notifies the provided students of new recommended internships.
- `selection_notification(Student student, Internship internship)`
Notifies the provided student that their CV has been selected for the provided internship.
- `selection_notification(Company company, CV CV)`
Notifies the provided company that a student has selected one of their internships.
- `match_notification(Company company, Match match)`
Notifies the provided company of a created match between one of its internships and a student.
- `sat_quest_notifications(Company company, Student student)`
Notifies company and student involved in a match about the availability of a satisfaction questionnaire.
- `match_notification(Student student, Match match)`
Notifies the provided student of a created match between their CV and an internship.
- `sel_quest_notification(Student student)`
Notifies the provided student about the availability of a selection process questionnaire.
- `interview_notification(Student student)`
Notifies the provided student about a new scheduled interview.
- `quest_comp_notification(Company company, Student student)`
Notifies the provided company that a student has compiled a questionnaire for the selection process regarding one of its internships.
- `admission_notification(Student student, Match active_int)`

Notifies the provided student about their admission to a matching internship.

- `rejection_notification(Student student, Match match)`

Notifies the provided student about their rejection for a matching internship.

- `mex_notification(Company company, Internship internship, Student student)`

Notifies the provided company about a message sent by the provided student in the context of an active internship.

- `mex_notification(Student student, Internship internship)`

Notifies the provided student of a message sent by a company in the context of an active internship.

DataAccessManager

- `reg_student(List<String> student_info)`

Triggers queries to the DBMS in order to register a student through the provided information. If the student e-mail is unique, it returns a success message (*String*) to the Authentication Manager, otherwise it throws an *Exception*.

- `reg_company(List<String> company_info)`

Triggers queries to the DBMS in order to register a company through the provided information. If the company e-mail is unique, it returns a success message (*String*) to the Authentication Manager, otherwise it throws an *Exception*.

- `check_credentials(List<String> credentials)`

Triggers a query to the DBMS in order to check the existence of the provided user credentials. If credentials are valid, it returns a success message (*String*) to the Authentication Manager, otherwise it throws an *Exception*.

- `upload_CV(CV CV, String userID)`

Triggers a query to the DBMS in order to upload a student CV. Returns a success message (*String*) to the Student Manager.

- `update_recomm(CV CV, List<String> keywords)`

Triggers queries to the DBMS in order to update recommendations (for the owner of the provided CV) with the given keywords. Returns an *Object* containing new recommendations (internships and related companies) for the corresponding student.

- `remove_CV(String CVID)`

Triggers a query to the DBMS in order to delete a student CV. Returns a success message (*String*) to the Student Manager.

- `remove_recomm_int(String CVID)`

Triggers a query to the DBMS in order to delete all recommendations associated to the specified CV. Returns a success message (*String*) to the Recommendation Manager.

- `pub_internship(String title, Internship document, String companyID)`

Triggers a query to the DBMS in order to upload an internship. Returns an *Internship* object to the Company Manager.

- `update_recomm(Internship internship, List<String> keywords)`

Triggers queries to the DBMS in order to update recommendations (for the publisher of the provided internship) with the given keywords. Returns an *Object* containing new recommendations (students) for the corresponding company.

- `get_internship(String internshipID)`

Triggers a query to the DBMS in order to retrieve all information related to a published internship (recommended CVs, interested students, completed matches...). Returns an *Object* containing the obtained data to the Company Manager.

- `remove_internship(String internshipID)`

Triggers a query to the DBMS in order to delete an internship. Returns a success message (*String*) to the Company Manager.

- `remove_recomm_CV(String internshipID)`

Triggers a query to the DBMS in order to delete all recommendations associated to the specified internship. Returns a success message (*String*) to the Recommendation Manager.

- `select_CV(String internshipID, String CVID)`

Triggers a query to the DBMS in order to add the specified CV to the "selected CVs" for a given internship. Returns an *Object* containing information about the related student and internship.

- `get_recom(String CVID)`

Triggers a query to the DBMS in order to retrieve all recommended internships for a given student (identified by the provided CVID). If there are any recommendations, it returns a *List<Internship>* object to the Student Manager, otherwise it throws an *Exception*.

- `select_int(String internshipID, String CVID)`

Triggers a query to the DBMS in order to add the specified internship to the "selected internships" for a given student (identified by the provided CVID). Returns an *Object* containing information about the related company and CV.

- `interested_companies(String CVID)`

Triggers a query to the DBMS in order to retrieve all internships for which companies have selected the provided CV. Returns a *List<Internship>* object to the Matching Manager.

- `accept_int(String internshipID, String CVID)`

Triggers a query to the DBMS in order to create a match between the specified CV and internship. Returns an *Object* containing information about the created match and the related company and student.

- `save_quest(List<Questionnaire> quests)`

Triggers a query to the DBMS in order to save the provided questionnaires. Returns a success message (*String*).

- `accept_CV(String internshipID, String CVID)`

Triggers a query to the DBMS in order to create a match between the specified CV and internship. Returns an *Object* containing information about the created match and the related company and student.

- `setup_interview(String matchID, Interview interview, Questionnaire quest)`

Triggers queries to the DBMS in order to correctly set up an interview with a student in the context of a match. If the specified date and time are free, it returns a *Student* object (related to the interviewed student) to the Selection Manager, otherwise it throws an *Exception*.

- `get_calendar(String studentID)`

Triggers a query to the DBMS in order to retrieve the set of scheduled interviews for the provided student. Returns a *List<Interview>* object to the Selection Manager.

- `update_rank(Questionnaire questionnaire)`

Triggers a query to the DBMS in order to update the suitability rank for the company that created the provided selection process questionnaire. Returns an *Object* containing information about student and company involved in the selection process.

- `admit_student(String matchID)`

Triggers a query to the DBMS in order to activate an internship program based on the provided match. Returns an *Object* containing the information about the activated internship and the involved student and company.

- `reject_student(String matchID)`

Triggers a query to the DBMS in order to reject a student for an internship program based on the provided match. Returns an *Object* containing information about the related match and the involved student.

- `get_sat_requests(String studentID)`

Triggers a query to the DBMS in order to retrieve the set of satisfaction questionnaires for the provided student. Returns a *List<Questionnaire>* object to the Recommendation Manager.

- `update_recomm(List<String> keywords, Questionnaire questionnaire)`

Triggers queries to the DBMS in order to update recommendations for internships and CVs that match the provided keywords. Recommendations are updated or not based on the questionnaire evaluation. The interface returns an *Object* containing information about students and internships (and related companies) whose recommendations have been modified.

- `get_stud_messages(String studentID)`

Triggers a query to the DBMS in order to retrieve the messaging sessions related to the provided student. Returns a *List<Message>* object to the Messaging Manager.

- `save_stud_mex(Message mex, String studentID, String internshipID)`

Triggers a query to the DBMS in order to update a messaging session with the provided student message. Returns an *Object* containing information about the related active internship and the involved student and company.

- `get_int_messages(String internshipID)`

Triggers a query to the DBMS in order to retrieve the messaging sessions related to the provided internship. Returns a *List<Message>* object to the Messaging Manager.

- `save_comp_mex(Message mex, String studentID, String internshipID)`

Triggers a query to the DBMS in order to update a messaging session with the provided company message. Returns an *Object* containing information about the related active internship and the involved student.

2.5.3. Data Layer

As mentioned in Section 2.4, interactions between `DataAccessManager` and DBMS are represented as simple method invocations, whereas they instead involve complex MongoDB query executions. Therefore, parameter types won't be specified for Data Layer interfaces, as they will all be converted into JSON structures in order to comply with MongoDB standards.

- `check_email(email)`

Finds the student or company associated to the provided e-mail address.

- `reg_student(student_info)`

Registers a new student by saving the provided information in the database.

- `reg_company(company_info)`

Registers a new company by saving the provided information in the database.

- `check_credentials(credentials)`

Finds the student or company associated to the provided user credentials.

- `upload_CV(CV, userID)`

Saves a new CV in the database and associates it to the student corresponding to the provided identifier.

- `find_similar_int(keywords)`

Finds internship documents that match the provided keywords.

- `update_recomm(CV, internshipID[ ])`

Updates recommendations for the provided CV with the given internships (and vice versa) and returns information about the latter and related companies.

- `remove_CV(CVID)`

Deletes the CV corresponding to the provided identifier from the database.

- `remove_recomm_int(CVID)`

Removes the CV corresponding to the provided identifier from all internships for which the first is recommended.

- `pub_internship(title, document, companyID)`

Saves a new internship in the database and associates it to the company corresponding to the provided identifier. Returns information about the the stored internship.

- `find_similar_CVs(keywords)`

Finds CV documents that match the provided keywords.

- `update_recomm(internship, CVID[ ])`

Updates recommendations for the provided internship with the given CVs (and vice versa) and returns information about the corresponding students.

- `get_internship(internshipID)`

Retrieves all information related to a published internship (recommended CVs, interested students, completed matches...).

- `remove_internship(internshipID)`

Deletes the internship corresponding to the provided identifier from the database.

- `remove_recomm_CV(internshipID)`

Removes the internship corresponding to the provided identifier from all CVs for which the first is recommended.

- `select_CV(internshipID, CVID)`

Adds the specified CV to the "selected CVs" for a given internship and retrieves information about the corresponding student and internship.

- `get_recom(CVID)`

Finds all recommended internships for the student associated with the provided CV identifier.

- `select_int(internshipID, CVID)`

Adds the specified internship to the "selected internships" for a given student (i.e. the one associated with the provided CV identifier) and retrieves information about the corresponding company and CV.

- `interested_companies(CVID)`

Finds all internships for which companies have selected the given CV.

- `accept_int(internshipID, CVID)`

Creates a match between the provided CV and internship (i.e. the ones associated to the specified identifiers). Retrieves information about the corresponding student and company and about the match creation.

- `save_quest(quests)`

Saves the provided questionnaires in the database.

- `accept_CV(internshipID, CVID)`

Creates a match between the provided internship and CV (i.e. the ones associated to the specified identifiers). Retrieves information about the corresponding company and student and about the match creation.

- `check_date(matchID, interview)`

Finds an interview (among the ones set up by the company involved in the provided match) whose date and time match those specified in the *interview* parameter.

- `setup_interview(matchID, interview, quest)`

Creates a new interview using information specified in the *interview* parameter and associates it with the provided match and questionnaire. Saves the provided questionnaire in the database. Retrieves information about the interviewed student.

- `get_calendar(studentID)`

Retrieves all information regarding the interviews in which the specified student (i.e. the one associated with the provided identifier) is involved.

- `update_rank(questionnaire)`

Updates the provided selection process questionnaire in the database with the assigned grade and retrieves information about the involved student and company.

- `admit_student(matchID)`

Activates an internship program based on the provided match and retrieves information about the activated internship and the involved student and company.

- `reject_student(matchID)`

Rejects a student for an internship program based on the provided match and retrieves information about the match and the involved student.

- `get_sat_quests(studentID)`

Retrieves all satisfaction questionnaires regarding matches and active internships in which the specified student is involved.

- `save_sat_quest(questionnaire)`

Updates the provided satisfaction questionnaire in the database with the assigned evaluation.

- `update_recomm(internshipID[ ],CVID[ ])`

Updates recommendations for the provided CVs with the given internships (and vice versa) and retrieves information about the corresponding companies, students and internships.

- `get_stud_messages(studentID)`

Retrieves all messaging sessions related to the specified student.

- `save_stud_mex(mex, studentID, internshipID)`

Using the provided student message, it updates the messaging session related to the active internship involving the specified student and internship. It also retrieves information about the corresponding student, company and internship.

- `get_int_messages(internshipID)`

Retrieves all messaging sessions related to the specified internship.

- `save_comp_mex(mex, studentID, internshipID)`

Using the provided company message, it updates the messaging session related to the active internship involving the specified student and internship. It also retrieves information about the corresponding student and internship.